Adaptive random reshuffling: a matching worst-case lower bound

Abstract

Consider random reshuffling of a smooth convex finite sum with n components over K epochs. We prove a matching worst-case lower bound when every update uses a positive scalar step at most $1/L$ and the output is any scalar convex combination of the actual iterates. Steps and output weights may depend on the instance, private randomness, and even the entire bank of sampled permutations. For every parameter tuple, one instance of dimension at most n is hard for all such choices. Its lower bound is a universal constant times

$$\min\left\{LD^2, \frac{LD^2}{nK} + \min\left\{\frac{\sigma D}{\sqrt{nK}}, \left(\frac{L\sigma^2 D^4}{nK^2}\right)^{1/3}\right\}\right\}.$$

The matching upper bound follows from Liu's constant-step, pre-update-average guarantee and the option of returning the initial point. The lower uses regular-simplex offsets, a squared-distance cone potential, and a pathwise interval estimate that pays for every auxiliary reset. A shortest-maximizing-interval argument controls the endpoint terms without imposing confinement on the actual trajectory.

1 Model and result

Fix integers $n \geq 2$, $K \geq 1$, put $T = nK$, and let $L > 0$ and $D, \sigma \geq 0$. The class $\mathcal{F}(n, L, D, \sigma)$ consists of instances in arbitrary finite-dimensional Euclidean spaces with differentiable convex components $f_1, \ldots, f_n$ such that

$$\|\nabla f_i(x) - \nabla f_i(y)\| \leq L\|x - y\|, \qquad f = \frac{1}{n}\sum_{i=1}^{n} f_i, \qquad x_* \in \arg\min f, \tag{1}$$

and specified initial point x_0 satisfying

$$\|x_0 - x_*\| \leq D, \qquad \frac{1}{n}\sum_{i=1}^{n}\|\nabla f_i(x_*)\|^2 \leq \sigma^2. \tag{2}$$

No global bound on gradient disagreement is imposed. Individual components need not be bounded below in the class definition.

Let $\omega = (\pi_1, \ldots, \pi_K)$ consist of independent uniform permutations of $\{1, \ldots, n\}$. If $t = en + j$ with $0 \leq e < K$ and $0 \leq j < n$, write $i_t = \pi_{e+1}(j + 1)$. We study the literal updates

$$x_{t+1} = x_t - \eta_t \nabla f_{i_t}(x_t), \qquad 0 < \eta_t \leq \frac{1}{L}, \tag{3}$$

and outputs

$$\widehat{x} = \sum_{t=0}^{T} w_t x_t, \qquad w_t \geq 0, \qquad \sum_{t=0}^{T} w_t = 1. \tag{4}$$

For the lower bound we allow η_t and w_t to be arbitrary measurable functions of the instance, ω, and an arbitrary private seed independent of ω. The constraints in (3)–(4) need hold only almost surely. In particular, the selection may use the whole permutation bank before the first update. This contains policies that see only the current epoch's permutation and choose a step after observing the current gradient. The choice of scalar step and convex output is unrestricted within this model; the update direction and sampled order remain those in (3). Expectations of the nonnegative loss are understood in the extended sense.

Define

$$\begin{aligned} A_0 &= LD^2, & B_0 &= \frac{LD^2}{T}, \\ N_1 &= \frac{\sigma D}{\sqrt{T}}, & N_2 &= \left(\frac{L\sigma^2 D^4}{nK^2}\right)^{1/3}, \\ N &= \min\{N_1, N_2\}, & \Psi &= \min\{A_0, B_0 + N\}. \end{aligned} \tag{5}$$

Theorem 1. *For each $n \geq 2$, $K \geq 1$, $L > 0$, and $D, \sigma \geq 0$, there is one instance in $\mathcal{F}(n, L, D, \sigma)$, of dimension at most n, such that every selection in (3)–(4) satisfies*

$$\mathbb{E}\big[f(\widehat{x}) - f(x_*)\big] \geq \frac{\Psi}{524288}. \tag{6}$$

The instance is chosen before the seed space, seed law, and step/output selection. Conversely, a parameter-dependent constant-step procedure, using either the uniform pre-update average or the initial point, satisfies the class-uniform upper bound 57Ψ.

The dimension bound concerns the hard witness, not the definition of $\mathcal{F}$. For any fixed private-seed probability space, let $\mathcal{P}$ be the class of instance-dependent selections above, and let $\mathcal{R}(I, P)$ denote their expected actual loss. The theorem gives

$$\frac{\Psi}{524288} \leq \sup_{I \in \mathcal{F}} \inf_{P \in \mathcal{P}} \mathcal{R}(I, P) \leq \inf_{P \in \mathcal{P}} \sup_{I \in \mathcal{F}} \mathcal{R}(I, P) \leq 57\Psi. \tag{7}$$

The lower is uniform even over the enlarged full-bank policy class. The upper uses one procedure from its much smaller constant-step subclass. Neither an upper for every policy nor equality of the two minimax values is asserted.

Relation to previous results. Liu [1] establishes the smooth-convex upper rate used here, including a theorem for steps that vary by epoch. The powers of the parameters in (5) are therefore inherited. Cha, Lee, and Yun [2] prove constant-step lower bounds for weighted epoch-iterate outputs, as well as results for chosen permutations. Rajput, Lee, and Papailiopoulos [3] give fixed-instance constant-step obstructions and discuss variable-step lower bounds. Thus fixed-instance constructions, weighted-output lower bounds, and the cubic noise rate are prior ingredients. The cone witness below itself has globally bounded gradient disagreement, $\|\nabla f_i(x) - \nabla f(x)\| = \sigma$; weakening that assumption is not a contribution of the construction. The claim here is a matching lower for adaptive scalar updates, all-state convex outputs, and all n, K and parameter regimes under a uniform smoothness bound L. It addresses the optimized-rate part of the lower-bound question in Liu's Section 5; it does not give the risk profile of every prescribed schedule, match the separate nonuniform-smoothness refinements, or settle every question posed there. It also does not cover momentum, vector-valued preconditioners, extra update directions, or outputs outside the convex hull of the iterates.

2 The inherited upper bound

Theorem 1 of Liu [1], with all component smoothness bounds set to L, implies that a constant $0 < \eta \le 1/(6L)$ and the *pre-update* average

$$\overline{x} = \frac{1}{T}\sum_{t=0}^{T-1} x_t \tag{8}$$

satisfy

$$\mathbb{E}[f(\overline{x}) - f(x_*)] \le \frac{6D^2}{\eta T} + 51\sigma^2 \min\{\eta, nL\eta^2\}. \tag{9}$$

This is precisely an admissible output in (4), with $w_T = 0$. In particular it is not an epoch-end average or a final-iterate guarantee.

For $D, \sigma > 0$, use either

$$\eta_1 = \min\left\{\frac{1}{6L}, \frac{D}{\sigma\sqrt{T}}\right\}, \qquad \eta_2 = \min\left\{\frac{1}{6L}, \left(\frac{D^2}{nL\sigma^2 T}\right)^{1/3}\right\}.$$

Since the reciprocal of a minimum is a maximum, (9) gives respectively $36B_0{+}57N_1$ and $36B_0{+}57N_2$. Choosing the smaller noise branch gives at most $57(B_0{+}N)$. Smoothness of the mean and $\nabla f(x_*) = 0$ also give $f(x_0) - f(x_*) \le A_0/2$. Choosing between this initial-point output and (8), using only the parameters, yields

$$\min\{A_0/2, 57(B_0 + N)\} \le 57\Psi.$$

When $\sigma = 0 < D$, use $\eta = 1/(6L)$ and the same initial-point option. When $D = 0$, return x_0 while taking any legal positive steps. These cases require no division by zero. This proves the upper part of Theorem 1 from the cited result. The remainder proves the lower independently of (9).

3 A legal cone instance in every component count

For now assume $D, \sigma > 0$. Set $\theta = 2^{-14}$ and choose $0 < \ell \le L$ with

$$\sigma^2 K \ge \ell^2 D^2 n. \tag{10}$$

The final choice of ℓ will cover every parameter regime. Put

$$\mu = \theta \min\left\{\ell, \frac{\sigma}{D\sqrt{T}}, \left(\frac{\ell\sigma^2}{nK^2D^2}\right)^{1/3}\right\}, \qquad a = \sqrt{\frac{\mu}{\ell}}, \tag{11}$$

$$\beta = \frac{\ell}{4(1+a^2)}, \qquad b = \beta a, \qquad H = \mu + ba. \tag{12}$$

3.1 Fixed regular-simplex blocks

For $n = 2$ take a single block of size two. For $n \ge 3$, write $n = 3q + j$, $j \in \{0, 1, 2\}$. If $j = 0$, take q blocks of size three. If $j = 1$ or $j = 2$, take respectively one block of size four or five and $q - 1$ blocks of size three. This includes $n = 3, 4, 5$ without an exceptional construction. If r is the number of blocks, then

$$r \le n/3 \quad (n \ge 3), \qquad r \le n/2 \quad (n \ge 2). \tag{13}$$

All blocks and label assignments are fixed before any permutations.

For a block of size m, use the zero-sum subspace of $\mathbb{R}^m$ and the vectors

$$\xi_i = \sqrt{\frac{m}{m-1}}\left(e_i - \frac{1}{m}\mathbf{1}\right), \qquad 1 \le i \le m. \tag{14}$$

They have norm one, sum zero, and satisfy $\langle \xi_i, \xi_j \rangle = -1/(m-1)$ for $i \neq j$. Different blocks occupy orthogonal subspaces. Their direct sum V has dimension $n - r$. The same notation ξ_i denotes the resulting fixed vectors indexed by the original n labels.

3.2 Objective and actual gradients

On the Euclidean product $\mathbb{R} \oplus V$, define the closed convex cone and potential

$$\begin{aligned} \mathcal{C}_a &= \{(u, v) : \|v\| \le -au\}, \\ \Phi(u, v) &= \frac{\mu}{2}u^2 + \frac{\ell}{8}\|v\|^2 + \frac{\ell}{8}\operatorname{dist}((u, v), \mathcal{C}_a)^2, \qquad (15) \\ f_i(u, v) &= \Phi(u, v) + \sigma\langle \xi_i, v \rangle, \qquad x_0 = (-D, 0). \qquad (16) \end{aligned}$$

The mean is exactly Φ, with minimizer $(0, 0)$. We recall the projection fact used to check global regularity. For a nonempty closed convex set C, its projection P_C satisfies $\langle x - P_C x, z - P_C x \rangle \le 0$ for $z \in C$. Applying this twice shows that $r(x) = x - P_C x$ is 1-Lipschitz. If $g(x) = \|r(x)\|^2/2$, comparison with the fixed competitors $P_C x$ and $P_C y$ gives

$$g(y) \le g(x) + \langle r(x), y - x \rangle + \tfrac{1}{2}\|y - x\|^2, \qquad g(y) \ge g(x) + \langle r(y), y - x \rangle - \tfrac{1}{2}\|y - x\|^2.$$

It follows that $\nabla g = r$. Convexity follows by using a convex combination of the two projected points as a feasible competitor. Since $\mu \le \theta\ell < \ell/4$, the quadratic part of (15) has gradient Lipschitz constant $\ell/4$. The distance term contributes another $\ell/4$. Thus all components are convex and globally $\ell/2$-smooth, hence L-smooth. Their gradients at zero are $(0, \sigma\xi_i)$, so the variance in (2) is exactly σ^2. The initial distance is D, and the dimension is $n - r + 1 \le n$. In fact these components are bounded below, although the public class does not require it.

Write a reachable point as $(-q, v)$, with $q > 0$ and $R = \|v\|$. If $R \le aq$, it is in the cone. Otherwise its projection is

$$\left(-\widehat{q}, \frac{a\widehat{q}}{R}v\right), \qquad \widehat{q} = \frac{q + aR}{1 + a^2}. \tag{17}$$

For completeness, at a fixed candidate slow coordinate $-s$, $s \ge 0$, the nearest fast point is the projection of v onto the ball of radius as. Minimizing $(s - q)^2 + (R - as)_+^2$ gives (17) in the outside case; here $0 < \widehat{q} < R/a$. This also verifies the formula at the boundary by continuity.

Consequently the *actual* update for current offset ξ is

$$\begin{aligned} q' &= (1 - \mu\eta)q + b\eta(R - aq)_+, \qquad v' = (1 - d\eta)v - \sigma\eta\xi, \qquad (18) \\ d &= \frac{\ell}{4} + \beta\frac{(R - aq)_+}{R}. \end{aligned}$$

The fraction is defined to be zero at $R = 0$. In particular,

$$\frac{\ell}{4} \le d \le \frac{\ell}{2}, \qquad \rho := 1 - d\eta \ge \frac{1}{2}, \qquad q' \ge (1 - \mu\eta)q > 0. \tag{19}$$

These inequalities use the original cap $\eta \le 1/L \le 1/\ell$. Starting from $q_0 = D$, they justify the reachable-state formulas at every update, for every legal selection.

Lemma 2 (Radius and energy)**.** *Every actual path satisfies $R_t \le 4\sigma/\ell$. On every interval $J = [s,t) \subseteq [0,T)$, with $S_J = \sum_{k\in J}\eta_k$ and $W_J = \sum_{k\in J}\eta_k R_k$,*

$$\sum_{k\in J}\eta_k^2 \le \frac{R_t^2 - R_s^2}{\sigma^2} + \frac{6W_J}{\sigma}. \tag{20}$$

Proof. The norm update gives $R' \le (1-\ell\eta/4)R + \sigma\eta$. The asserted radius follows by induction from $R_0 = 0$. Moreover,

$$\begin{aligned} R'^2 - R^2 &= -(2d\eta - d^2\eta^2)R^2 - 2\rho\sigma\eta\langle v,\xi\rangle + \sigma^2\eta^2 \\ &\ge \sigma^2\eta^2 - \ell\eta R^2 - 2\sigma\eta R \ge \sigma^2\eta^2 - 6\sigma\eta R. \end{aligned}$$

Sum this inequality over the actual interval. No conditional expectation or independence of the step and label is used. □

4 Dual tables from the sampled permutations

Fix the entire permutation bank. In one epoch, a block of size m occupies positions $p_1 < \cdots < p_m$ in $\{0,\dots,n-1\}$. Choose a largest consecutive gap $g = p_{k+1} - p_k$, with deterministic tie-breaking, and let E be its k earlier labels. Set $z = \sum_{i\in E}\xi_i$. The fixed simplex identities give

$$\langle z,\xi_i\rangle = \begin{cases} (m-k)/(m-1), & i \in E, \\ -k/(m-1), & i \notin E, \end{cases} \qquad \|z\|^2 = \frac{k(m-k)}{m-1}.$$

Use dual vectors

$$\phi^- = \frac{m-1}{m-k}z, \qquad \phi^+ = -\frac{m-1}{k}z. \tag{21}$$

Keep ϕ^- up to p_k, interpolate linearly to ϕ^+ across the gap, and keep ϕ^+ thereafter. There is no hit from this block inside the chosen gap. Hence the dual has inner product one with each actual offset at its hit position. Also

$$\|\phi^-\|^2 = \frac{(m-1)k}{m-k}, \quad \|\phi^+\|^2 = \frac{(m-1)(m-k)}{k}, \quad \|\phi^+ - \phi^-\| = m\sqrt{\frac{m-1}{k(m-k)}} \le m.$$

For $2 \le m \le 5$, the endpoint norms are at most 4 and the change is at most 5.

Taking the orthogonal direct sum over all blocks gives a dual ϕ_t at every update position such that

$$\langle \phi_t, \xi_{i_t}\rangle = 1, \qquad \|\phi_t\| \le 4\sqrt{r}, \qquad |\langle \phi_t,\xi_i\rangle| \le 4 \quad \text{for every original label } i. \tag{22}$$

The last bound holds even when the label and the dual time differ. This fact is needed when paying for resets.

Write $g_{e,j}$ for the selected gap of block j in epoch e, and define

$$B_e = \sum_{j=1}^{r}\frac{1}{g_{e,j}}, \qquad G_A(\omega) = A\left[8K\sqrt{r} + 5\sqrt{n-1}\sum_{e=1}^{K}\sqrt{B_e}\right]. \tag{23}$$

Within an epoch, the squared variation of block j is at most $25/g_{e,j}$. Orthogonality and Cauchy–Schwarz therefore give

$$\sum_{\text{internal } t}\|\phi_{t+1} - \phi_t\|^2 \le 25B_e, \qquad \sum_{\text{internal } t}\|\phi_{t+1} - \phi_t\| \le 5\sqrt{(n-1)B_e}. \tag{24}$$

An epoch-boundary jump has norm at most $8\sqrt{r}$. Thus for any interval $[s,t)$ and any vectors $z_s,\dots,z_t$ bounded in norm by A, summation by parts gives

$$\left|\sum_{k=s}^{t-1}\langle \phi_k, z_k - z_{k+1}\rangle\right| \le G_A(\omega). \tag{25}$$

Indeed the two endpoints cost at most $8A\sqrt{r}$, at most $K-1$ boundary jumps cost $8A(K-1)\sqrt{r}$, and the internal variations are bounded by (24). This is simultaneous over all intervals and all such vector sequences.

Lemma 3 (Expected geometry)**.** *For every $n \ge 2$, $\mathbb{E}B_e \le 4$, and for deterministic $A \ge 0$,*

$$\mathbb{E}G_A \le 16AK\sqrt{n}. \tag{26}$$

Proof. For $n \ge 3$, fix any three labels of a block. Their positions form a uniform three-element subset: each position subset is realized by $3!(n-3)!$ of the $n!$ permutations. If its span is s, there are $(n-s)(s-1)$ such subsets, so

$$\mathbb{E}\frac{1}{s} = \binom{n}{3}^{-1}\sum_{s=2}^{n-1}\frac{(n-s)(s-1)}{s} \le \binom{n}{3}^{-1}\frac{(n-1)(n-2)}{2} = \frac{3}{n}.$$

The outer span of the whole block is at least s, and its largest gap is at least its outer span divided by $m-1 \le 4$. Thus $\mathbb{E}(1/g_{e,j}) \le 12/n$. By (13), $\mathbb{E}B_e \le 4$. For $n=2$ there is one gap, equal to one. Finally Jensen gives $\mathbb{E}\sqrt{B_e} \le 2$, and

$$\mathbb{E}G_A \le AK\big(8\sqrt{n/2} + 10\sqrt{n-1}\big) < 16AK\sqrt{n}$$

when $A > 0$. The non-strict version also covers $A = 0$. Only the marginal positions of fixed labels were used. The duals may be correlated with every action of the policy. □

5 Paying for excursions of the actual trajectory

The next estimate is pathwise. Fix a bank, a legal realized trajectory, an interval $J=[s,t)$, and deterministic $h, A > 0$ with $A > \sigma h$. Keep update k if

$$\eta_k < h, \qquad R_k \le A, \qquad R_{k+1} \le A, \tag{27}$$

and remove the other updates. Subscripts keep and rem denote the corresponding partial sums.

Lemma 4 (Paid auxiliary resets)**.** *For every such interval,*

$$S_{\text{keep}} \le \frac{G_A}{\sigma} + 4S_{\text{rem}} + \frac{2\ell\sqrt{r}}{\sigma}W_J + \frac{4\sqrt{r}}{\sigma}R_s. \tag{28}$$

Proof. Construct an auxiliary sequence solely for accounting. Set $z_s = v_s$ if $R_s \le A$, and $z_s = 0$ otherwise. Copy the actual path on kept updates. During each maximal removed run freeze the auxiliary sequence until the run's final update. If the actual state at the run's endpoint has radius at most A, reset there to that state. Otherwise leave it frozen. Such an unsafe endpoint can occur only on a terminal removed run: a following kept update would require that same state to be safe. Thus the construction agrees with v at both endpoints of every kept update and remains within radius A at every time.

For a removed run ending safely, the reset displacement is the sum of the actual fast increments along the run. The sole possible additional term is v_s, when the first run starts at an unsafe initial state. Test the reset against the dual at its final update. For every update inside that run, (18) and the all-label bound in (22) give

$$|\langle \phi_j, v_{k+1} - v_k \rangle| \le 4\sigma\eta_k + 2\ell\sqrt{r}\,\eta_k R_k \quad \text{for every test time } j.$$

Consequently all reset charges together are at most

$$4\sigma S_{\mathrm{rem}} + 2\ell\sqrt{r}\,W_{\mathrm{rem}} + 4\sqrt{r}\,R_s. \tag{29}$$

Every removed actual increment is charged at most once. On a kept update, (22) yields the exact identity

$$\sigma\eta_k = \langle \phi_k, z_k - z_{k+1} \rangle - d_k\eta_k\langle \phi_k, v_k \rangle.$$

Sum over kept updates, insert the fully paid resets, and use (25). The kept drift terms cost at most $2\ell\sqrt{r}\,W_{\mathrm{keep}}$. Combining with (29) and dividing by σ proves (28).

This auxiliary sequence is not a modified algorithm. In particular no actual state is reset or projected, no excursion is discarded without payment, and a removed run may cross epoch boundaries. The argument works for intervals and keep/remove choices selected after the full trajectory has been seen. □

5.1 An unrestricted interval estimate

Set $C = HD$ and choose

$$b_0 = b - \frac{2C\ell\sqrt{r}}{\sigma}, \qquad h = \frac{120C}{b_0\sigma}, \qquad A = \frac{130C}{b_0}. \tag{30}$$

We first check that these expressions are positive. From (10) and the linear cap in (11),

$$a^2 \le \theta, \qquad a^2 \le \frac{\theta\sigma^2}{\ell^2 D^2 n}, \qquad \frac{H}{b} = a(5 + 4a^2).$$

Together with $r \le n/2$, these give

$$\frac{2C\ell\sqrt{r}}{\sigma b} \le 2(5 + 4\theta)\sqrt{\theta/2} < \frac{1}{16}.$$

For the last inequality, use $5 + 4\theta < 21/4$ and $\sqrt{\theta} = 1/128$, obtaining $21/(256\sqrt{2}) < 1/16$. In particular,

$$\frac{15b}{16} \le b_0 \le b, \qquad A - \sigma h = \frac{10C}{b_0} > 0. \tag{31}$$

Lemma 5 (All intervals)**.** *With* (30)*, every actual interval $J = [s, t)$ satisfies*

$$CS_J - bW_J \le \frac{CG_A}{\sigma} + \frac{4C\sqrt{r}}{\sigma}R_s + \frac{b_0}{24\sigma}R_t^2. \tag{32}$$

Proof. Add S_{rem} to (28), multiply by C, and subtract bW_J. The result is

$$CS_J - bW_J \le \frac{CG_A}{\sigma} + \frac{4C\sqrt{r}}{\sigma}R_s + 5CS_{\mathrm{rem}} - b_0 W_J. \tag{33}$$

Every large update, meaning $\eta_k \geq h$, is removed. By Lemma 2,

$$\begin{aligned} 5CS_{\text{large}} &\leq \frac{5C}{h}\sum_{k\in J}\eta_k^2 \\ &\leq \frac{b_0}{24\sigma}(R_t^2 - R_s^2) + \frac{b_0}{4}W_J. \end{aligned} \tag{34}$$

This is valid even if $h > 1/L$ and there are no large updates: the right-hand bracket before multiplying by $5C/h$ is nonnegative by (20).

A small removed update has at least one unsafe endpoint. Since $R_{k+1} \leq R_k + \sigma\eta_k$, its pre-update radius satisfies $R_k > A - \sigma h = 10C/b_0$. Therefore

$$5CS_{\text{small,rem}} \leq \frac{b_0}{2}W_{\text{small,rem}}. \tag{35}$$

Insert (34)–(35) into (33). The remaining weighted-radius terms are nonpositive, as is the starting-energy term. Dropping them proves (32). □

6 Endpoint trimming and a bank-only certificate

For an actual path define its maximal interval deficit

$$Z_{\text{path}} = \max\left\{0, \max_{0\leq s<t\leq T}\sum_{k=s}^{t-1}\eta_k(C - bR_k)\right\}. \tag{36}$$

The global radius bound alone is too costly in the endpoint terms of (32). The following finite selection argument retains small endpoints instead.

Lemma 6 (Endpoint trim)**.** *Every legal path for a fixed bank satisfies*

$$Z_{\text{path}} \leq \mathcal{Z}(\omega) := \frac{C}{L} + \frac{CG_A(\omega)}{\sigma} + \frac{4C^2\sqrt{r}}{\sigma b} + \frac{C^2}{24\sigma b}. \tag{37}$$

The certificate $\mathcal{Z}$ is nonnegative and depends only on the parameters and permutation bank, not on the steps or output.

Proof. If $Z_{\text{path}} > 0$, choose a shortest interval attaining it. Its first and last increments are strictly positive: a negative endpoint could be deleted to increase the maximum, while a zero endpoint could be deleted without changing it. As all steps are positive, $R_s < C/b$ and $R_{t-1} < C/b$.

Delete the final update, whose contribution is at most C/L. If the remainder is empty, this already bounds the maximum. Otherwise apply (32) to $[s, t-1)$. Its initial and final state radii are both less than C/b. Using $b_0 \leq b$ gives the three remaining terms of (37). The case $Z_{\text{path}} = 0$ is immediate. This argument conditions on no selected endpoint event. □

Lemma 7 (Mean certificate)**.** *The construction satisfies*

$$\mathbb{E}\mathcal{Z} < \frac{D}{128}. \tag{38}$$

Proof. By (31), $A \le 2080C/(15b)$. Use Lemma 3, $\sqrt{r} \le K\sqrt{n}$, and $1 \le K\sqrt{n}$ in (37) to obtain

$$\begin{aligned}\frac{\mathbb{E}\mathcal{Z}}{D} &\le \frac{H}{L} + \left(\frac{33280}{15} + 4 + \frac{1}{24}\right)\frac{H^2 DK\sqrt{n}}{\sigma b} \\ &= \frac{H}{L} + \frac{53345}{24}\frac{H^2 DK\sqrt{n}}{\sigma b}. \end{aligned} \tag{39}$$

The coefficients in (12) obey

$$H \le \frac{5}{4}\mu, \qquad \frac{H^2}{b} \le \frac{25}{4}(1+\theta)\frac{\mu^{3/2}}{\sqrt{\ell}} < 7\frac{\mu^{3/2}}{\sqrt{\ell}}. \tag{40}$$

The linear and cubic caps defining μ therefore imply

$$\frac{\mathbb{E}\mathcal{Z}}{D} < \frac{5}{4}\theta + 7\theta^{3/2}\frac{53345}{24} < \frac{1}{128}.$$

For a finite check, $53345/24 < 2240$, so the middle expression is less than $(160 + 15680)/2^{21} < 1/128$. □

7 Retention of every state and adaptive output

For a pre-state with $q_k \le D$, (18) gives

$$q_{k+1} \ge q_k - C\eta_k + b\eta_k R_k. \tag{41}$$

Indeed $(R - aq)_+ \ge R - aq$ and $C = (\mu + ba)D$.

Proposition 8. *For the one cone instance above, every allowed selection satisfies*

$$\mathbb{E}\Phi(\widehat{x}) \ge \frac{\mu D^2}{16}. \tag{42}$$

Proof. Suppose an actual state first falls below $D/2$ at time t. If there is no earlier state with $q_j > D$, put $s = 0$. Otherwise let $j < t$ be the last such time and put $s = j + 1$. In this second case, (19) implies $q_s > (1 - \mu/L)D$; in the first case $q_s = D$. Since $\mu/L \le \theta < 1/4$, we have $s < t$. All pre-states in $[s, t)$ satisfy $q_k \le D$. Summing (41) shows

$$Z_{\text{path}} \ge \sum_{k=s}^{t-1} \eta_k(C - bR_k) \ge q_s - q_t > (1/2 - \mu/L)D > D/4.$$

By Lemma 6, on every bank with $\mathcal{Z}(\omega) \le D/4$ *all* legal selections have $q_t \ge D/2$ for every $0 \le t \le T$. This event is independent of the choice of policy in the sense that its definition involves only the bank; no statistical independence of actions is needed. Markov's inequality and (38) give

$$\mathbb{P}\{\mathcal{Z} > D/4\} < 1/32.$$

Every scalar convex output on the complementary event has slow coordinate $-\widehat{q}$ with $\widehat{q} \ge D/2$, even when the weights use the entire trajectory and seed. Since $\Phi(u, v) \ge \mu u^2/2$ and $\Phi \ge 0$ everywhere,

$$\mathbb{E}\Phi(\widehat{x}) \ge \frac{31}{32}\frac{\mu D^2}{8} \ge \frac{\mu D^2}{16}.$$

For product-almost-everywhere legal selectors, the same argument applies outside their null set. Continuity of the actual gradients and measurable selectors makes the finitely many states and the loss measurable. Nonnegative integration requires no additional integrability assumption. The instance and its good-bank event are unchanged for every independent private seed law. □

8 All regimes and the single-instance choice

For arbitrary $D, \sigma > 0$, choose

$$\ell = \min\left\{L, \frac{\sigma\sqrt{K}}{D\sqrt{n}}\right\}. \tag{43}$$

This is a smoothness scale of the constructed instance; the algorithm still obeys the original cap $1/L$. It ensures (10). Let $M = \min\{A_0, N_1, N_2\}$. If $\ell = L$, (11) immediately gives $\mu D^2 = \theta M$. If $\ell < L$, put $s_0 = \sigma/(D\sqrt{T})$. Then $\ell = Ks_0$, and the cubic term in (11) is exactly s_0. Thus $\mu = \theta s_0$. Moreover $\sigma^2 K < L^2 D^2 n$ implies $N_1 < A_0$ and $N_1 < N_2$. We conclude in both cases that

$$\mu D^2 = \theta M, \qquad \mathbb{E}\Phi(\widehat{x}) \geq \frac{M}{262144}. \tag{44}$$

The other candidate is the identical scalar quadratic

$$f_i(x) = \frac{\lambda}{2}x^2, \qquad \lambda = \frac{L}{2T}, \qquad x_0 = D, \qquad x_* = 0. \tag{45}$$

It is legal for every $\sigma \geq 0$, since its optimum variance is zero. Along every allowed realized schedule,

$$x_t = D\prod_{j=0}^{t-1}(1 - \lambda\eta_j) \geq D\left(1 - \sum_{j=0}^{t-1}\lambda\eta_j\right) \geq D/2.$$

Here every factor lies in $[0, 1]$, and the product bound follows inductively. Any scalar convex output therefore satisfies

$$f(\widehat{x}) - f(0) \geq \frac{\lambda D^2}{8} = \frac{B_0}{16}. \tag{46}$$

Choose between (16) and (45) by comparing the two proved lower bounds, before any policy or randomness is chosen. Since $B_0 + M \geq \Psi$,

$$\max\left\{\frac{M}{262144}, \frac{B_0}{16}\right\} \geq \frac{B_0 + M}{524288} \geq \frac{\Psi}{524288}.$$

This is one parameter-only witness, not an instance selected after observing the strategy. For $\sigma = 0 < D$, use only the quadratic; then $\Psi = B_0$. For $D = 0$, $\Psi = 0$, the quadratic is again legal, and returning x_0 achieves zero. These branches avoid all undefined cone parameters. The proof of Theorem 1 is complete.

9 Verification and scope

The lower theorem has also been expressed in Lean 4 using Mathlib. Its final statement is `Gate149.FinalTheorem.one_fixed_instance_fresh_epoch_lower`. It constructs an actual standard-Euclidean instance of dimension at most n before quantifying over the seed space, probability law, and measurable full-bank selectors. The formal development includes the globally convex smooth objective, its actual gradient recursion, fixed-label simplex geometry, paid resets, interval trimming, parameter tuning, and the joint probability law. In particular, the finite uniform permutation-bank law is proved equal to the product of fresh uniform epoch laws. The final lower statement has been accepted by the Lean kernel, with its axiom report containing only `propext`, `Classical.choice`, and `Quot.sound`.

The formal lower and the inherited upper have different validation scopes. The proof of Liu's published upper theorem is not included in that development. Its minimax sandwich is an explicitly conditional formal corollary given the upper strategy, while Section 2 supplies the mathematical citation and tuning. No claim about priority or external assessment follows from kernel acceptance.

The substantive lower-bound mechanism is the simultaneous pathwise charge (28)–(32), followed by endpoint trimming. It prevents full-order information, adaptive scalar steps, and trajectory-selected convex averaging from improving the optimized worst-case rate on this class. This is a statement about the specified scalar-gradient model, not a limitation on all finite-sum optimization methods or a prediction of practical training performance.

References

[1] Zijian Liu. Random reshuffling dominates stochastic gradient descent. In *Proceedings of the 39th Conference on Learning Theory*, volume 336 of *Proceedings of Machine Learning Research*, pages 4859–4882, 2026. `https://proceedings.mlr.press/v336/liu26d.html`.

[2] Jaeyoung Cha, Jaewook Lee, and Chulhee Yun. Tighter lower bounds for shuffling SGD: Random permutations and beyond. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 3855–3912, 2023. `https://proceedings.mlr.press/v202/cha23a.html`.

[3] Shashank Rajput, Kangwook Lee, and Dimitris Papailiopoulos. Permutation-based SGD: Is random optimal? In *International Conference on Learning Representations*, 2022. `https://openreview.net/forum?id=YiBa9HKTyXE`. Full version: `https://arxiv.org/abs/2102.09718v2`.